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Department of Computer Science & Engineering

Subject: Theory of Computation
Subject Code: 23CSI602

Module 1: Introduction

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FINITE AUTOMATA

Finite automata are basic modules of a computer system,which addresses simple real time problems.Figure 1.1 shows conceptual view of finite automata. It has an input tape, read head and finite controller. Input tape is divided into finite cells and each cell stores one input symbols. Read head is used to read the input symbol from the input tape. The finite controller used to record the next state of finite automata.

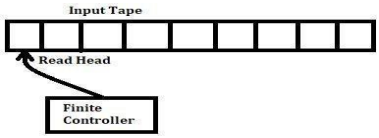


Figure1.1:Conceptual view of finite automata

Before discussing finite automata in detail, it is very much important to know the basic requirement for the finite automata. They are

i. Sets.
ii. Graphs.
iii. Languages.
iv. Starclosure.
v. Plusclosure.

Sets (S):

Asetiscollectionofelements,eachelementhasitsownidentity.Asetcanberepresentedby enclosingits elements in curlybraces. for example, theset of small letters a, b, c is shownas

S={a,b,c}

Set{2,4, 6,…}denotesthe setofall positiveevenintegers.we can usemore explicitnotation,in which we write

S={i|iis> 0andiseven}

Wecanperformfollowingoperationsonsets,they are

a. union(U),
b. intersection(∩),and
c. difference(−)
d. complementationdefinedas

$S_1 U S_2 = \{x \mid x \in S_1 or x \in S_2\}$ $S_1 \Omega$

$S_2 = \{x \mid x \in S_1 and\ x \in S_2\}$

$S_1 - S_2 = \{x \mid x \epsilon S_1 and x \notin S_2\}$

Thecomplement of a set S, denoted byṠ consists of all elements not inS. To make this meaningful, we needto knowwhat the universal set Uof all possible elements is. If U is specified, then

$\dot{S} = \{x \mid x \epsilon U, x \notin S\}$

Theset withnoelements, calledtheemptyset or thenullset,is denotedbyε. Fromthedefinitionofaset, it is obvious that

SUε=S -ε=S

SΩε=ε ε̈=

U

DeMorgan'slaws: Thefollowingusefulidentities,knownasDeMorgan's laws.

$\overline{S_1 \cup S_2} = S_1 \cap S_2$

$\overline{S_1 \cap S_2} = S_1 \cup S_2$

areneededonseveraloccasions.

SubsetandProperSubset: Aset $S_1$is saidtobeasubsetofS,everyelement ofS_1isalsoanelement of S.Wewritethis as $S_1 \subseteq S$.If$S_1 \subseteq S$,butScontains anelement not inS_1, wesaythat $S_1$is aproper subset of S; we write this as $S_1 \subset S$.

DisjointSets: IfS_1andS_2 havenocommon element, that is,$S_1 \cap S_2 = \varepsilon$ thenthesets aresaidtobe disjoint.

FiniteandInfinitesets: Asetissaidtobefiniteifitcontainsa finitenumber of elements;otherwiseitis infinite. Thesize of a finiteset is the number of elements init; this is denoted by |S|.

Power-Set: Agivenset hasmanysubsets.Thesetofallsubsets iscalledthepower-set andis denotedby 2^S. Observe that 2^S is a set of sets.

IfSistheset{a,b,c},thenitspowersetis

2^S={ε,{a},{b},{c},{a,b},{b,c},{a,c},{a,b.c}} Here |S| = 3

and $|2^S|$ = 8.

Cartesianproduct: Orderedsequences ofelements fromother sets aresaidtobetheCartesianproduct. For the Cartesianproduct of two sets, which itself is a set of orderedpairs, we write

$S=S_1 x S_2$

LetS1= {2,4}andS2={2,3,5,6}.ThenS1×S2= {(2,2),(2,3),(2,5),(2,6),(4,2),(4,3),(4,5),(4, 6)}. Notethattheorderinwhichtheelements ofa pairarewritten matters.Thepair (4,2)isinS1×S2, but (2, 4) is not.

Graphs:

Agraphisaconstructconsistingoftwofinitesets,thesetV={υ1,υ2,…,υn}ofverticesandthesetE= {e1, e2,…,em}ofedges.Eachedgeisapairofvertices fromV,forinstance, $e_i=(v,v_j)$.

Languages:

A finite, nonempty set of symbols are called the alphabet and is represented with symbol ∑. Fromthe individual symbols weconstruct strings, which arefinitesequences of symbols fromthealphabet. For example, ifthealphabet∑ ={0},then∑*todenotetheset ofstrings obtainedbyconcatenatingzeroor more symbols from ∑.

Wecanwrite∑*={ε,0,00,000…}

If∑={0,1}
0and1areinputalphabetsorsymbols,then:
(0^2=00,0^3=000,1^2=11,1^3=111,whereasinmathematicstheyarenumbers0^2=0,1^2=1)
L=∑*={ε,0,1,00,01,10,11,000,001,010 .....}

Kleenclosure(*closure):
*closure={$\sum^0 U \sum^1 U \sum^2 U \sum^3$ .....}
$\sum^0$={ε} $\sum^1$={0,1}$\sum^2$={00,01,10, 11}
$\sum^3$={000,001,010,011,100,101,110,111}
Therefore∑*={ε,0,1,00,01,10,11,000,001,010,011,100,101,……}
$^+$Closureon∑={0,1}
$\sum^+$={$\sum^1 U \sum^2 U \sum^3$…}
∑*=$\sum^+$+ε

Therefore$\sum^+$=∑*-ε

FormalLanguagesandAutomataTheory:ITisamathematicalmodelusedtodesigncomputerprograms andsequentiallogiccircuits.Itisstatemachineswhichcapturesandmakestransitionto acceptorreject. Finiteautomata has input tape, read head and finitecontroller **Input tape**: It is divided into finitecell and eachcellisusedtostoreoneinputsymbol. **Readhead**:Readheadisusedtoreadinputsymbolfrominput tape. **Finite Controller**: Finite controller records, the transition to next state. The default value of finite controller is initial state q_0. States arerepresented with circles; they are of three types,

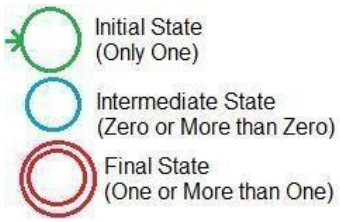


Figure1.2:Differenttypesofstatesusedinfiniteautomata

Statesareconnected bylinescalledtransition.Ithastwoends,one endiscalledinitiationpointandthe other end is called termination point (>).

Initiation⟶Termination

Thecombinationoftransitionsandstatesiscalledatransitiondiagram.

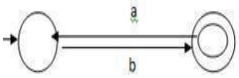


Figure1.3:Atypicaltransitiondiagram

Figure1.4showsatransitiondiagramtomapcitiesandtheir roads.

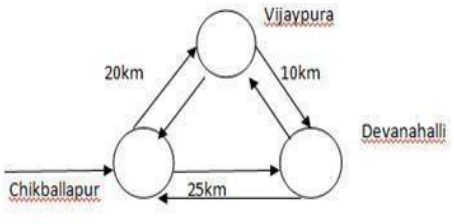


Figure 1.4: Finite automata with real time example.

FiniteAutomataareclassifiedintotwotypes.Theyare:
1) DeterministicFiniteAutomata(DFA)
2) NonDeterministicFiniteAutomata(NFA)
InDeterministicFiniteAutomatanumberofoutgoingtransitionsarewelldefinedfromeachstatedepending uponinput symbols. For example, ifthereis oneInput symbol, thenyoufindoneoutgoing transitionfrom each state, for two input symbols two outgoingtransitions are present and so on.

Example1.1: DrawtheDFA,theinputsymbolis∑= {0}

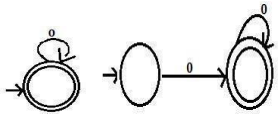


Figure1.5:DFA'sforinputsymbol∑={0}.

Example1.2: DrawtheDFA,theinputsymbolis ∑={0,1}

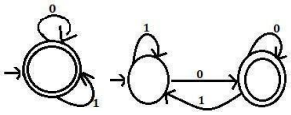


Figure1.6:DFA'sforinputsymbol∑={0,1}.

Definition of DFA: we define DFA as,

M_{DFA}=(Q,∑,δ,Initialstate,Finalstate)

Where,Qissetofstates,∑ =setofinputsymbols,δ: transitionfunctionis From the $Qx\Sigma \rightarrow Q$

table 1.

Q={q_0,q_1,q_2}∑={a,b,c}

δ:
δ(q_0,a)=q_1 δ(q_0,b)=q_0 δ(q_0,c)=q_2
δ(q_1,a)=q_1 δ(q_1,b)=q_2 δ(q_1,c)=q_0
δ(q_2,a)=q_2 δ(q_2,b)=q_1 δ(q_2,c)=q_2
Note:InitialStateis$q_0$andfinalstateisq_2

Table1.1:Transitiontable

Q\∑	a	b	c
→q_0	q_1	q_0	q_2
q_1	q_1	q_2	q_0
*q_2	q_2	q_1	q_2

NondeterministicFiniteAutomata(NFA):

Ifoutgoingtransitionsarenotdependentonnumberofinputsymbols,thensuch automataarecalledas Nondeterministic Finite Automata.

Forexample:∑={0, 1}

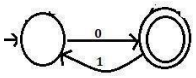


Figure1.7:NondeterministicFiniteAutomatafor∑={0,1}

Thereisnooutgoingtransitionfrominitialstate,forasymbol‘1‘andsimilarlythereisnooutgoing transition from final state for ‘0’.

TransitionFunctionforstring:

Note:1)δisthetransitionfunction,usedforsymbols.

2)Forthestring(Setofsymbols), $\hat{\delta}$ is used

{}=ε

{a}=a

$\hat{\delta}$ (**q_0**,ε)=**q_0//q_0** isreadingempty string, string isempty therefore,transition to same state **q_0**.

We already Proved $\hat{\delta}(q_0,\varepsilon) = q_0$

$\hat{\delta}$ (**q_0**,a)=$\hat{\delta}$ (**q_0**,εa)=δ($\hat{\delta}$ (**q_0**,ε),a)=δ(,a) Therefore

$\hat{\delta}$ (**q_0**,a)=δ (**q_0**,a)

Thereforeforsinglesymbolstring $\hat{\delta}$ =δ

$\hat{\delta}$ (**q_0**,wa)

Where‘w’isthemulti-symbolstringand‘a’isasymbol,henceweseparatethe transitionfunctionfor strings and symbols as given below:

δ($\hat{\delta}$ (**q_0**,w),a)

Example1.3: Writeatransitiontableforadfashowninfigure1.8. DFA over

alphabet ∑ = {a, b}

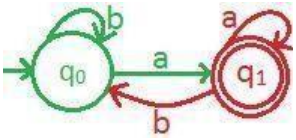


Figure 1.8: Transition diagram o0f a DFA.

Table1.2:Transitiontableforthefigure1.8

Q \ ∑	a	b
→q0	δ(q0,a)	δ(q0, b)
*q1	δ(q1, a)	δ(q1,b)

Q \ ∑	a	b
→q0	q1	q0
*q1	q1	q0

M_{DFA} =(Q,∑,δ,IS,FS) Q
={ q_0, q_1}
∑={a,b}
δ (q_0, a) =q_1S δ q_0,b)
=q_0δ(q_1,a)=q_1δ(q_1,b)=q_0IS
=q_0FS={q_0}

Example1.4: Constructtransitiontableforthediagramshowninfigure1.9,overthealphabet∑={0,1,2}

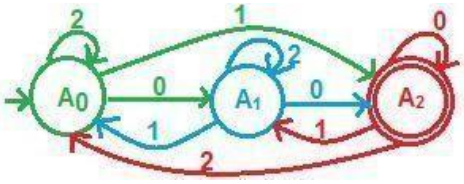


Figure1.9:Transitiondiagram

Table1.3:Transitiontableforthefigure1.9

A \ ∑	0	1	2
→A0	A1	A2	A0
A1	A2	A0	A1
*A2	A2	A1	A0

WecandefineaboveDFAas:M_{DFA}=(A,∑,δ,IS,{FS}) δ:

δ (A_0, 0)=A_1

δ(A_0,1)=A_2δ(A_0,2)=A_0δ(

A_1,0)=A_2 δ(A_1,1)=A_0δ(A_1,2)=A_1

$\delta(A_2,0)=A_2$ $\delta(A_2,1)=A_1\delta(A_2,2)=A_0$

$A=\{A_0,A_1,A_2\};\sum=\{0,1,2\};IS=A_0;FS=A_2$

String Acceptance:

The first symbol of string begins with initial state. Later, traverses zero or more than zero intermediate state(s)for theintermediatesymbol(s).Finallylastsymbolofthe stringendswith finalstate ofautomata. This is called string accepted or otherwise string is not accepted.

Example1.5: Find outwhether thegivenstringisacceptedbythe DFA.Assume thestringasabab. From the given string ∑ = {a,b} . Note that always string is traversed by left side to right side,

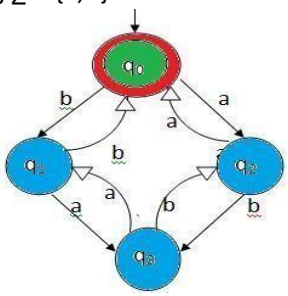


Figure1.10:Transitiondiagramforexample1.5.

$\hat{\delta}(q_0,a)=\delta(q_0,a)=q_2$ —— 1

$\hat{\delta}(q_0,ab)=\delta(\hat{\delta}(q_0,a)b)$ from 1 $\hat{\delta}(q_0,a)=q_2$

$=\delta(q_2,b)=q_3$ —— 2

$\hat{\delta}(q_0,aba)=\delta(\hat{\delta}(q_0,ab),a)$ **fromequation2** $\hat{\delta}(q_0,ab)=q_3$

$=\delta(q_3,a)=q_1$

$\hat{\delta}(q_0,abab)=\delta(\hat{\delta}(q_0,aba),b)$

$=\delta(\hat{\delta}(q_1,b)=$

Find out whether given string 10110 is accepted by DFA.

From given string ∑={0,1}

10110

$(q_0, 1)=\delta(q_0, 1)=q_2$ ------- ①

10110

$(q_0, 10)=\delta(\ (q_0, 1),0)=\delta(q_2, 0)=q_2$ ------- ②

10110 $(q_0, 101)=(\ (q_0, 10),1)=\delta(q_2, 1)=q_1$

10110 $(q_0, 1011)=\delta(\ (q_0, 101),1)=\delta(q_1, 1)=q_2$

10110 $(q_0, 10110)=\delta(\ (q_0, 1011),0)=\delta(q_2, 1)=q_2$

q2 is a final state hence given string 10110 is accepted.

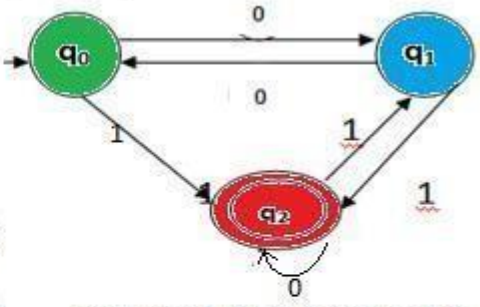


Figure 1.11 : Transition diagram

q_0Givenstringisacceptedbya DFA.

Example1.6: Findoutthelanguageacceptedbythe DFA.

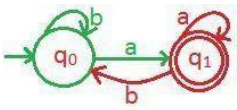


Figure1.12:TransitiondiagramofExample1.6.

$\delta(q_0,a)=q_1$; $\hat{\delta}(q_0,aa)=\delta(\hat{\delta}(q_0,a),a)=\delta(q_1,a)=q_1$ L={a,

aa, ba, baa, aba, …..}

Ingeneralwecanwrite:

L={w|wisastringendingwithsymbol a}

Example1.7: FindoutthelanguageacceptedbytheDFA.

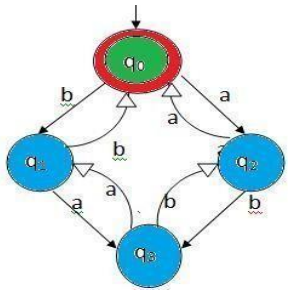


Figure1.13:TransitiondiagramofExample1.7

Initialstateisalsoafinalstate,henceitacceptemptystringε. $\hat{\delta}$

$(q_0,\varepsilon)=q_0$;

$\hat{\delta}(q_0,aa)=\delta(\hat{\delta}(q_0,a)a)=\delta(q_2,a)=q_0$;

$\hat{\delta}(q_0, bb)= \delta(\hat{\delta}(q_0, b)b)= \delta(q_1, b)=q_0$;

ThereforelanguageL={ε,aa,bb,aabb,abab…}

IngeneralwecanwriteL={w|wisastringconsistofevennumberofa'sandevennumberofb's}

DesignofDFA:

Example1.8:*DesignaDFAwhichacceptallthestringsendswithsymbolaoverthealphabet∑={a,b}.*

L={a,aa,ba,aba,bba…}

i. DrawtheNFAforthesmalleststringaofthe language. The NFA for the string a is

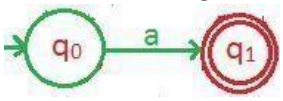


Figure1.14:NFAforthestringa

Convert all states of NFA to DFA, that is from each state two outgoing transitions, one for symbol 'a' anotherfor'b'.Frominitialstateonlyoneoutgoingtransitiononsymbol'a',onemoreoutgoingtransition isrequiredthatison'b'.Iftheconditioninthegivenproblemendswith,theoutgoing

transitionontheothersymbolb,definitelylooptoainitial state.

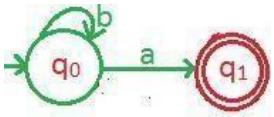


Figure1.14:secondoutgoingtransitionfromstateq_0

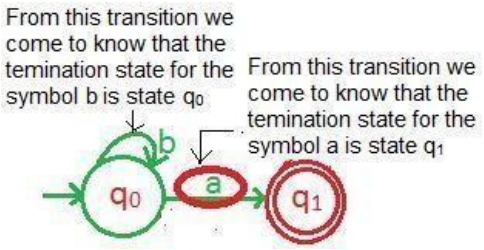
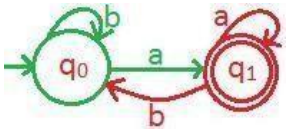


Figure1.15:$q_0$isterminationforsymbolband$q_1$isterminationforsymbola

Terminationstateforsymbol'a'is$q_1$andfor'b'itisq_0.Outgoingtransitionfrom$q_1$on symbol 'a' is loop, and for 'b' it is to q_0 .



Figyre1.16:DFAwhichacceptallthestringsendswitha. We can

define above DFA as M_{DFA}=(Q, ∑, δ , IS, {FS})
Q={q_0, q_1}
∑={a,b}
δ(q_0,a)=q_1δ(q_0,b)=q_0δ
(q_1,a)=q_1δ(q_1,b)=q_0IS
=q_0FS={q_1}

Example1.8: DesignaDFAwhichacceptallthestringsendswithsubstring01overthealphabet

∑={0,1}.

L={01,001,101,0001,1001…}.

i. DrawtheNFAforthesmalleststring01ofthelanguage. The NFA for the string 01 is

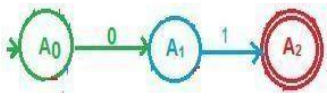


Figure1.17:NFAforthestring01.

ii. Convert all states of NFA to DFA, that is from each state two outgoing transitions, one for symbol'0'another for '1'. From initial state only one outgoing transition on symbol '0',one

more outgoing transition is required that is on '1'. If the condition in the given problem is, ends with,the outgoing transition on the other symbol 1,definitely loopto a initial state.

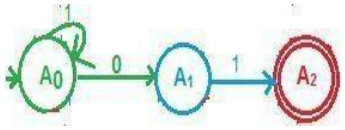


Figure1.18:conversionofNFAstateA_0toDFA

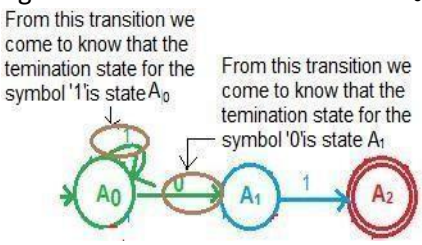
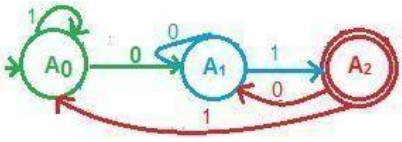


Figure1.19:A_0isterminationforsymbol1andA_1isterminationforsymbol0

Terminationstateforsymbol'0'isA_1andfor'1'itisA_0.OutgoingtransitionfromA_1on symbol '0' is loop. From A_2on symbol '0' to $A_1$and on symboll '1' to A_0.



Figyre1.20:DFAwhichacceptallthestringsendswith01. We

can define above DFA as M_{DFA}=(A, ∑, δ, IS, {FS})

A={A_0,A_1,A_2};∑={0, 1}

δ:
δ(A_0,0)=A_1δ(A_0,1)=A_0 δ (A_1,
0)=A_1δ (A_1, 1)=A_2 δ (A_2,
0)=A_1δ (A_2, 1)=A_0IS=A_0;
FS=A_2

Example1.9: DesignaDFAwhichacceptallthestringsendswithsubstring012overthealphabet

∑={0,1,2}.

L={012,0012,1012,00012,10012…}.

i. DrawtheNFAforthesmalleststring012ofthelanguage. The NFA for the string 012 is

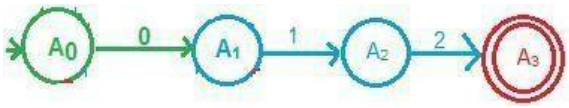


Figure1.21:NFAforthestring012.

ii. Convert all states of NFA to DFA, that is from each state three outgoing transitions, one for symbol '0' second for '1' third from '2'. From initial state only one outgoing transition on symbol '0', two more outgoing transitions are required that ison '1' and '2'. If the condition

in the given problem ends with, the outgoing transition on the other symbols '1'and '2', definitely loop to a initial state.

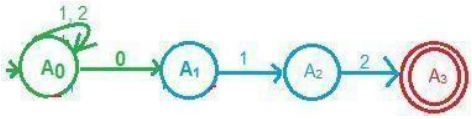


Figure1.22:conversionofNFAstateA_0toDFAstate.

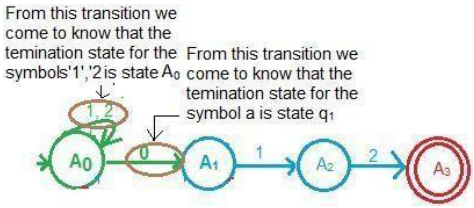


Figure1.23:A_0isterminationforsymbol1,2and$A_1$is terminationforsymbol0

Termination state for symbol '0'is $A_1$and for '1'and '2'is A_0. Outgoing transition from A_1on symbol '0' is loop and for'2' is A_0. From A_2on symbol '0' to $A_1$and on symbol '1' to A_0. From A_3on symbol '0' to $A_1$and on symbols '1' and '2' to A_0.

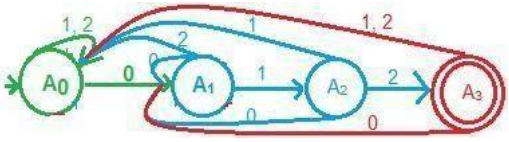


Figure1.24:DFAwhichacceptallthestringsendswith012. We

can define above DFA as M_{DFA}=(A, ∑, δ, IS, {FS})

A={A_0,A_1,A_2,A_3};∑={0,1,2}

δ:

δ(A_0,0)=A_1	δ(A_0,1)=A_0	δ(A_0,2)=A_0
δ(A_1,0)=A_1	δ(A_1,1)=A_2	δ(A_1,2)=A_0
δ(A_2,0)=A_1	δ(A_2,1)=A_0	δ(A_2,2)=A_3
δ(A_3,0)=A_1	δ(A_3,1)=A_0	δ(A_3,2)=A_0

IS=A_0; FS=A_3

Example1.10: DesignaDFAwhichacceptallthestringsbeginswithsymbol'a'overthealphabet

∑={a, b}.

L={a,aa,ab,abb,aba,aaa…}

i. DrawtheNFAforthesmalleststringaofthe language. The NFA for the string a is

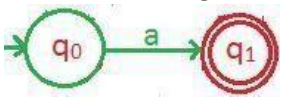


Figure1.25:NFAforthestringa.

Convert all states of NFA to DFA, that is from each state two outgoing transitions, one for symbol 'a' another for 'b'. From initial state only one outgoing transition on symbol 'a', one more outgoing transition is required that is on 'b'. If the condition in the given problem is, beginwith,theoutgoingtransitionon theother symbol 'b',istonewstate calleddeadstate. ***Dead state***: In finite automata the string travers from initial state to final state, in some situation we make finite automata to die called dead state. Dead state is a non-final state, from which all outgoing transitions are loop to the same state.

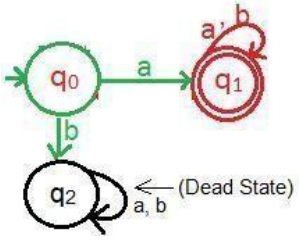


Figure1.26:DFAwhichacceptallthestringbeginwitha.

ii. Conditionisbegin,outgoingtransitionsfromfinalstateisloop.

WecandefineaboveDFAas M_{DFA} =(Q,∑,δ ,IS,{FS}) Q ={
q_0, q_1, q_2}
∑={a,b}
δ(q_0,a)=q_1δ(q_0,b)=q_2 δ
(q_1, a) =q_1δ (q_1, b) =q_1 δ
(q_2,a)=q_2δ(q_2,b)=q_2IS
=q_0FS={q_1}

Example1.11: DesignaDFAwhichacceptallthestringsbeginswithsubstring01overthealphabet

∑={0,1}.

L={01,010,011,0100,0101…}.

i. DrawtheNFAforthesmalleststring01ofthelanguage. The NFA for the string 01 is

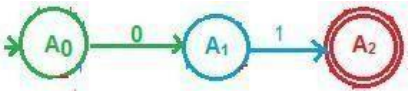


Figure1.27:NFAforthestring01.

ii. Convert all states of NFA to DFA, that is from each state two outgoing transitions, one for symbol '0' another for '1'. From initial and intermediate states only one outgoing transition, one more outgoing transitionis required. If the condition in the given problemis,beginwith, the outgoing transition from both the states to new state called dead state.

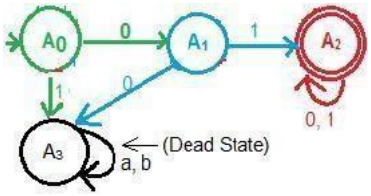


Figure1.28:DFAwhichacceptallthestringbeginwith01.

iii. Conditionisbegin,outgoingtransitionsfrom finalstateA_2onsymbol'0'and'1'isloop. We can define above DFA as M_{DFA}=(A, ∑, δ, IS, {FS})

A={A_0,A_1,A_2A_3};∑={0, 1}

δ:
δ (A_0, 0)=A_1 δ(A_0,1)=A_0δ (A_1, 0)=A_1 δ(A_1,1)=A_2δ (A_2,0)=A_1δ(A_2,1)=A_0 δ (A_3, 0)=A_3δ (A_3, 1)=A_3IS=A_0; FS=A_2

Example1.12: Designa DFAwhich accept allthe strings beginswith01orends with 01 orboth overthe alphabet ∑={0, 1}.

L={01,010,011,0100,001,101,0001,1001,0101,01101…}

WehavealreadydesignedboththeDFAs,joiningoftheseDFAs,isthesolutionforthegivenproblem.

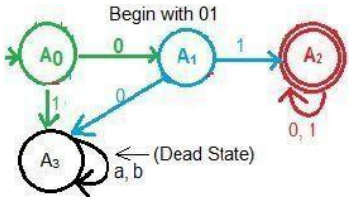


Figure1.29:DFAwhichacceptallthestringbeginwith01.

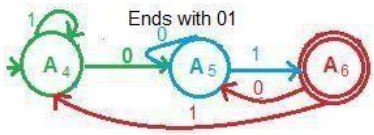


Figure1.30:DFAwhichacceptallthestringendswith01.

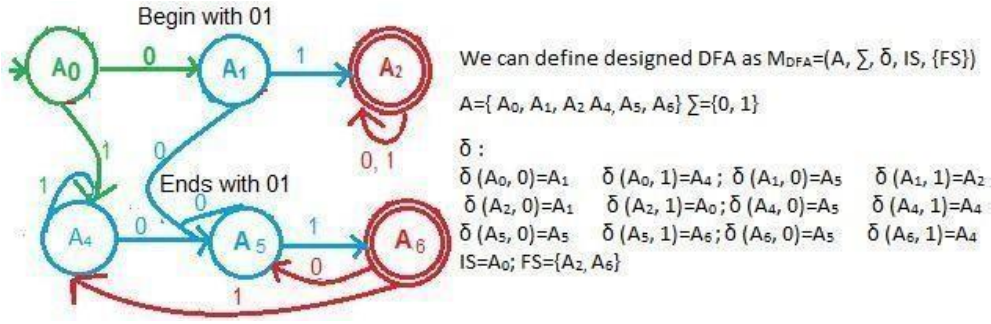


Figure1.31:DFAwhichacceptallthestringbeginwith01,endwith01beginandendwith01.

Example1.13: DesignaDFAwhichacceptallthebinarystringsdivisibleby2. L={ ε,

10, 100, 110, …}

TodesignDFAweneedtomakethefollowingassumptions.

$q_0 \rightarrow 0 \rightarrow q_0$

$q_0 \rightarrow 1 \rightarrow q_1$

$q_0 \rightarrow 10 \rightarrow q_0$

$q_0 \rightarrow 11 \rightarrow q_1$

$q_0 \rightarrow 100 \rightarrow q_0$

$q_0 \rightarrow 101 \rightarrow q_1$

Fromtheaboveassumption,wecandrawthefollowingDFA.

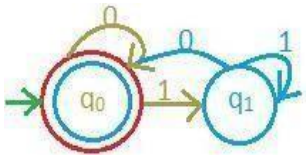


Figure1.32:DFAwhichacceptsallthestringsdivisibleby2. We can

define above DFA as M_{DFA}=(Q, ∑, δ , IS, {FS})
Q={q_0, q_1}
∑={0,1}
δ(q_0,0)=q_0δ(q_0,1)=q_1δ
(q_1,0)=q_0δ(q_1,1)=q_1IS
=q_0FS={q_0}
Example:

Example1.14: DesignaDFAwhichacceptallthebinarystringsdivisibleby5. L={ ε,

101, 1010, 1111, …}

To design DFA we need to make the following assumptions.

$q_0 \rightarrow 0 \rightarrow q_0$

$q_0 \rightarrow 1 \rightarrow q_1$

$q_0 \rightarrow 10 \rightarrow q_2$

$q_0 \rightarrow 11 \rightarrow q_3$

$q_0 \rightarrow 100 \rightarrow q_4$

$q_0 \rightarrow 101 \rightarrow q_0$

$q_0 \rightarrow 110 \rightarrow q_1$

$q_0 \rightarrow 111 \rightarrow q_2$

$q_0 \rightarrow 1000 \rightarrow q_3$

$q_0 \rightarrow 1001 \rightarrow q_4$

$q_0 \rightarrow 1010 \rightarrow q_5$

According to assumption we can draw the following DFA

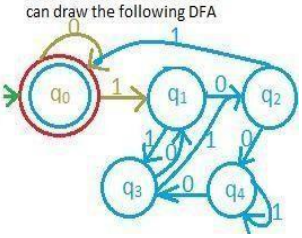


Figure1.33:DFAwhichacceptallthestringsdivisibleby5.

Wecandefine,designedDFAasM$_{DFA}$=(A,∑,δ,IS,{FS}) A={

q_0, q_1, q_2q_3, q_4} ∑={0, 1}

δ:
δ(q_0,0)=q_0 δ(q_0,1)=q_1;δ(q_1,0)=q_2 δ(q_1,1)=q_3
δ(q_2,0)=q_4 δ(q_2,1)=q_0;δ(q_3,0)=q_1 δ(q_3,1)=q_2
δ(q_4,0)=q_3 δ(q_4,1)=q_4;

IS=q_0;FS=q_0

Example1.15: DesignaDFAwhichacceptstringoverx,y everyblockoflength3 containsatleastonex. To design a DFA we need to make following assumptions,

1. Ifthereisno'x'inthestring,suchastringshouldnotbeaccepted.

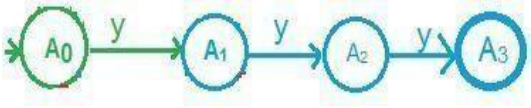


Figure1.34:NFAwhichisnotacceptingthreey'sstring.

2. One'x'inthestring,acceptedbyfiniteautomata.

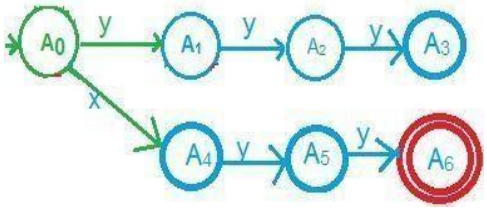


Figure1.35:NFAwhichisacceptingthreesymbolstringinwhichonesymbolisx.

3. ThepositionofA$_1$andA$_5$issameasA$_0$andA4.InbetweenA$_0$andA$_4$thesymbol is'x', inbetwwen A$_1$and A$_5$is also x and so on.

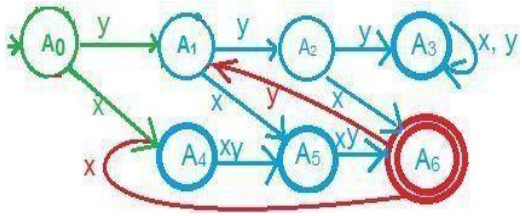


Figure1.36:*DFAwhichacceptstringoverx,yifeveryblockoflength3containsatleastonex.*

WecandefinedesignedDFAasM$_{DFA}$=(A,∑,δ,IS,{FS}) A= {

A_0, A_1, A_2, A_4,A_4, A_5, A_6} ∑={x, y}

δ:
δ(A_0,x)=A_4 δ(A_0,y)=A_1;δ(A_1,x)=A_5 δ(A_1,y)=A_2
δ(A_2,x)=A_6 δ(A_2,y)=A_3;δ(A_3,x)=A_3 δ(A_3,y)=A_3
δ(A_4,x)=A_5 δ(A_3,y)=A5;δ(A_5,x)=A_6 δ(A_5,y)=A_6;

$\delta(A_6, x)=A_4$

$\delta(A_6,y)=A_1I$

$S=A_0$; $FS=\{A_6\}$

Example1.16: DesignaDFAwhichacceptevennumberofrand evennumberofsoverthesymbol ∑={r, s}.

i. Draw separate DFA, one accept even number of r's and another one accept even number of s's.

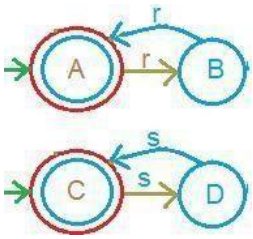


Figure1.37:*DFA'sacceptevensymbolsstringoverr,s.*

ii. Join two DFA,we get four new states AB, AC,BC,BD.Designed DFAaccepteven number of r andevennumberof s,outof four states ABsatisfiedthiscondition,hence thisstate isafinal state.

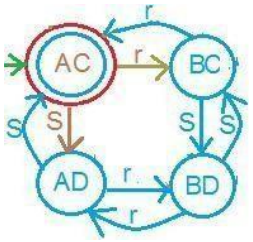


Figure 1.38: *DFA which accept even number of r and s.*

WecandefinedesignedDFAasM_{DFA}=(State,∑,δ,IS,{FS})

State={ AC, AD, BC, BD} ∑={r, s}

δ:
δ(AC,r)=BC δ(AC,s)= AD;δ(BC,r)=AC δ(BC,s)=BD
δ (AD, r)=BD δ (AD, s)=BD; δ (BD, r)=AD δ(BD,s)=BC

IS=AC; FS={AC}

Example1.17: DesignaDFAwhichacceptallthestringsdescribedbythelanguageL={$N_0(w)\geq1$and $N_1(w)=2$}

L={011, 110, 0011, 1010, 11100, 10101...}

i. Exchange I symbol of firststring with the lastsymbol of the firststring,the string obtained is the second string of the language.
ii. DrawNFAfortwosmalleststrings,forbothNFAinitialandfinalstatesaresame.

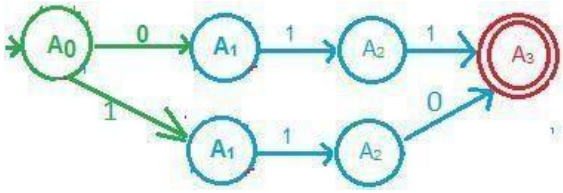


Figure1.39:TwoDFAsforthestring011and110withsameinitialandfinal state.

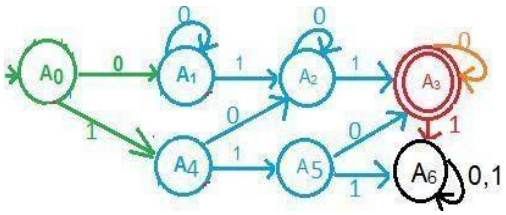


Figure1.40:DFAwhichacceptallthestringsdescribedbythelanguageL={$N_0(w)\geq 1$and$N_1(w)=2$}. We can

define designed DFA as M_{DFA}=(A, ∑, δ, IS, {FS})

A={A_0,A_1,A_2A_3,A_4,A_5,A_6} ∑={0,1}

δ:

δ(A_0,0)=A_1 δ(A_0,1)=A_4;δ(A_1, 0)=A_0 δ(A_1,1)=A_2

δ(A_2,0)=A_2 δ(A_2,1)=A_3;δ(A_4,0)=A_2 δ(A_4,1)=A_5

δ(A_5,0)=A_3 δ(A_5,1)=A_6;δ(A_6,0)=A_6 δ(A_6,1)=A_6

IS=A_0;FS={A_3}

NondeterministicFiniteAutomata(NFA)

In DFA the outgoing transitions are defined in advance,where as in NFA the outgoing transitions are not defined in advance. In DFA, the string is either accepted or not accepted, where as in NFA both are possible. The finite automata shown in the figure 1.41 is a NFA. There is no outgoing transition for the symbol 'a' from the state q_1.

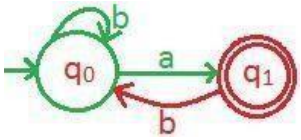


Figure1.41:NondeterministicFiniteAutomata

WecandefineaboveNFAasM_{DFA} =(Q,∑,δ ,IS,{FS}) Q ={

q_0, q_1,}

∑={a,b}

δ:

$Q^*{}_\Sigma \rightarrow 2^Q$

δ(q_0,a)=q_1δ(q_0,b)=q_0δ (q_1,

b) =q_0

δ(q_2,a)=q_2δ(q_2,b)=q_2

IS=q_0FS={q_1}

TheNFAshowninfigure1.42is,acceptingaswellasnotacceptingthestringhod.

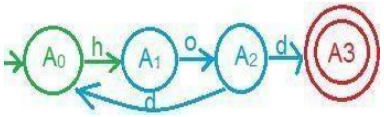


Figure1.42:Non-deterministicfiniteautomata

Example1.18: DesignNFAwhichacceptallstrings,ifsecondsymbolfromrightsideis1overthe alphabet ∑={0, 1}.

L={10,110,010,0110…}

i. Firststepdrawthefinalstate.

Figure1.43:Finalstate

ii. Drawtheintermediatestateandtransitionforthesymbols‘0’and‘1’.

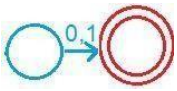


Figure1.44:Intermediateandfinalstates

iii. Drawtheinitialstateandtransitionforthesymbol‘1’.

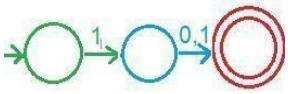


Figure1.45:Non-deterministicfiniteautomataacceptingtwostrings10,11.

iv. DesignedNFAhastoacceptall thestrings,if second symbol fromright sideissymbol‘1’, to do this, mark loop to initial state on symbols ‘0’ and ‘1’.

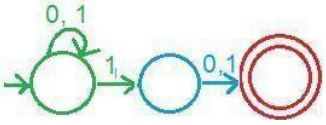


Figure1.46:NFAwhichacceptsallstrings,ifsecondsymbolfromrightsideis1overthealphabet∑={0,1}.

v. Namethestatesandstop.

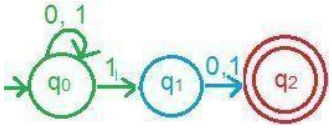


Figure1.47:NFAwhichacceptallstrings,ifsecondsymbolfromrightsideis1overthealphabet∑={0,1}.

DefinitionofNFA:NFAcanbedefinedas

M_{DFA}=(Q,∑,δ,IS,{FS}) Q
={ q_0, q_1, q_2}
∑={0,1}
δ:
δ(q_0,0)=q_0δ(q_0,1)={q_0,q_1} δ
(q_1, 0) =q_2
δ(q_1,1)=q_2
IS=q_0FS={q_2}

EquivalenceofNFAandDFA:

We have already seen both NFA and DFA, both are same in the definition but they differ in transition function. If number of states in NFA has two states $q_0$and q_1. Here, $q_0$is initial state and $q_1$is final state. The equivalent states in DFA's are *q_0], [q_1] and [q_0, q_1]. Where $q_0$is initial state and (q_1), (q_0,q_1) are the final states. The pair (q_0,q_1) is final state because; one of its states is final state. They are represented as shown in figure 1.48

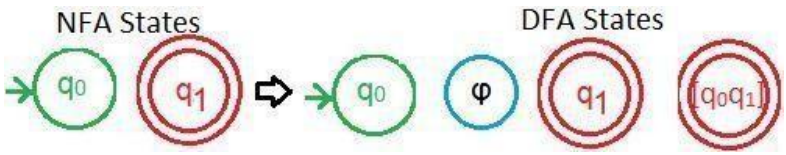


Figure1.48:EquivalencestatesofNFAandDFA.

Example1.19: ConverttheNFAshowninfigure1.49intoitsequivalentDFA.

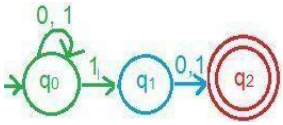


Figure1.49:Non-deterministicfiniteautomataof Example1.19.

The conversion begins with the initial state of the NFA which is also an initial state of DFA. Conversion generatesnumberofnewstates,foreverynewstate wearefindingtheoutgoingtransitionsfordifferent symbols.

Table1.4:DifferenttablesfortheconversionofNFAtoDFA.

Q \ Σ	0	1
→q_0	q_0	[q_0 q_1]

New state

Q \ Σ	0	1
→q_0	q_0	[q_0 q_1]
[q_0q_1]	[q_0q_2]	[$q_0q_1q_2$]

New state

New states

δ([q_0q_1], 0)=δ(q_0, 0)Uδ(q_1, 0)
=q_0 Uq_2
=q_0q_2
δ([q_0q_1], 1)=δ(q_0, 1)Uδ(q_1,1)
=q_0 q_1Uq_2
=$q_0q_1q_2$

Q \ Σ	0	1
→q0	q0	[q0 q1]
[q0q1]	q0q2	q0q1q2
q0q2	q0	[q0q1]

$\delta([q_0q_2], 0)=\delta(q_0, 0)\cup\delta(q_2, 0)$
$=q_0 \cup \emptyset$
$=q_0$
$\delta([q_0q_2], 1)=\delta(q_0, 1)\cup\delta(q_2, 1)$
$=q_0 q_1 \cup \emptyset$
$=q_0q_1$

Q \ Σ	0	1
→q0	q0	[q0 q1]
[q0q1]	q0q2	q0q1q2
[q0q2]	q0	[q0q1]
[q0q1q2]	[q0q2]	[q0q1q2]

$\delta([q_0q_1q_2],0)=\delta(q_0, 0)\cup\delta(q_1, 0)\cup\delta(q_2, 0)$
$=q_0 \cup q_2 \cup \emptyset$
$=q_0q_2$
$\delta([q_0q_1q_2],1)=\delta(q_0, 1)\cup\delta(q_1, 1)\cup\delta(q_2, 1)$
$=q_0 q_1 \cup q_2 \cup \emptyset$
$=q_0q_1q_2$

No further new states; mark the final state of DFA. Any pair of state which consists of final state of the NFA, the particular pair is a final state in DFA.

Q \ Σ	0	1
→q0	q0	[q0 q1]
[q0q1]	q0q2	q0q1q2
*[q0q2]	q0	[q0q1]
*[q0q1q2]	[q0q2]	[q0q1q2]

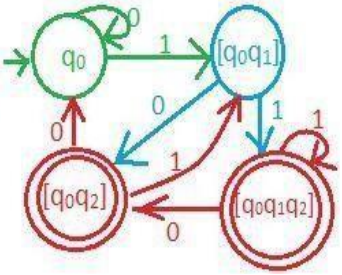


Figure1.50:EquivalenceDFAoffigure1.49

ApplicationofFiniteautomata:

Finiteautomatisusedtosearchatext.Therearetwomethodsusedtosearchtext,they are:

i. NondeterministicFiniteAutomata(NFA)
ii. DeterministicFiniteAutomata(DFA)

Tosearcha text“shine”and“hire”the NFAshowninfigure1.51isused The

input alphabets are ∑={s, h, i, n, e, r }

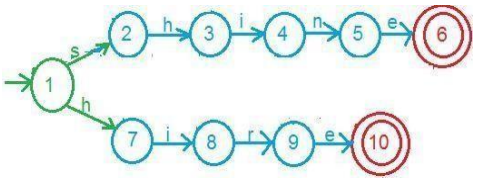


Figure1.51:NFAusedtosearchstringsshineandhire

Tosearchatext“shine”and“hire”theDFAshowninfigure1.52isused

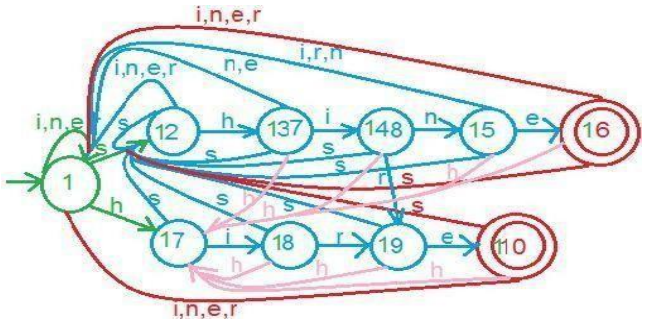


Figure1.52:DFAusedtosearchstringsendingwithshineandhire

Example1.20: Designafiniteautomata,thatreadsstringmadeupoflettersHONDA,and recognize those string that contain the word HOD as a substring.

i. FindoutthealphabetfromthestringHONDA.Theinputsymbolsare∑={H,O,N,D,A}
ii. DrawtheNFAforthesubstringHOD.

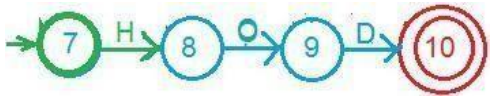


Figure1.53:NFAusedtosearchstringHOD

iii. ConvertNFAstateoffigure1.53intoDFAstates.

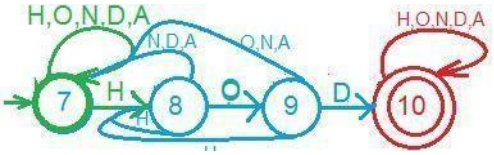


Figure1.54:DFAacceptallthestringsendswithHODoverthesymbols{H,O,N,D, A}

NFAwithε-transition:

Tillnowwhateverautomatawestudiedwerecharacterizedbyatransitionon thesymbols.Butthereare some finite automata having transition for empty string also, such NFAs are called NFA- ε transition.

ε-closure(q_0):stateswhichareclosed/relatedtostate$q_0$onemptytransition(ε)iscalledε-closure(q_0).

InDFAandNFA $\hat{\delta}(q_0,\varepsilon)=q_0$,whereasinε-NFA

i. $\hat{\delta}(q,\varepsilon)=\varepsilon\text{-closure}(\hat{\delta}(q,\varepsilon))=\varepsilon\text{-closure}(q)$.
ii. $\hat{\delta}(q,x)=\varepsilon\text{-closure}(\delta(\hat{\delta}(q,\varepsilon),x)=\varepsilon\text{-closure}(\delta(\varepsilon\text{-closure}(q),x)$. $\hat{\delta} \neq \delta$
iii. $\hat{\delta}(q,wx)=\varepsilon\text{-closure}(\delta(\hat{\delta}(q,w),x)$.

Example1.21: Findoutε-closureofallthestatesofε-NFAshowninfigure1.55.

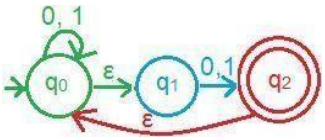


Figure1.55:NFAwithε-transitionofExample1.21.

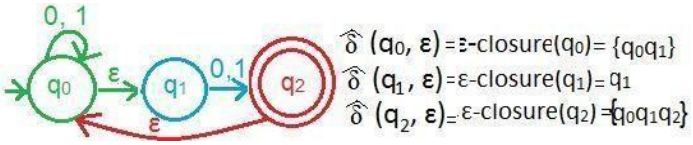


Figure1.56:ε-closureofallthestatesoffigure1.55

Example1.22: Convertthe ε-NFAshowninfigure1.57intoNFA

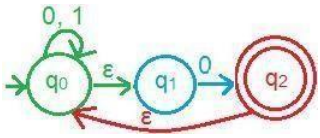


Figure1.57:ε-NFAofExample1.22.

Firstfindoutthe ε-closureofallthestates.

Note:state ε-closureconsistoffinalstateof ε-NFA,theparticularstateisafinalstateinNFA.

$\hat{\delta}(q_0, \varepsilon)$ =ε-closure(q_0)= {q_0q_1}
$\hat{\delta}(q_1, \varepsilon)$ =ε-closure(q_1)= q_1
$\hat{\delta}(q_2, \varepsilon)$= ε-closure($q_2$) ={$q_0q_1q_2$}

$\hat{\delta}$(q_0,0)=ε-closure(δ($\hat{\delta}$(q_0,ε),0).

=ε-closure(δ({q_0q_1},0).

=ε-closure(δ(q_0,0)Uδ(q_1,0)).

=ε-closure(q_0Uq_2).

=ε-closure(q_0)Uε-closure(q_2).

={q_0q_1}U{$q_0q_1q_2$}

={$q_0q_1q_2$}

$\hat{\delta}$(q_0,1)=ε-closure(δ($\hat{\delta}$(q_0,ε),1).

=ε-closure(δ({q_0q_1},1).

=ε-closure(δ(q_0,1)Uδ(q_1,1)).

=ε-closure($q_0U\phi$).

=ε-closure(q_0)Uε-closure(ϕ).

=$\{q_0q_1\}$

$\hat{\delta}(q_1,0)$=ε-closure($\delta(\hat{\delta}(q_1,\varepsilon),0)$.

=ε-closure($\delta(q_1,0)$).

=ε-closure(q_2).

=$\{q_0q_1q_2\}$

$\hat{\delta}(q_1,1)$=ε-closure($\delta(\hat{\delta}(q_1,\varepsilon),1)$.

=ε-closure($\delta(q_1,1)$).

=ε-closure(ϕ).

=ϕ

$\hat{\delta}(q_2,0)$=ε-closure($\delta(\hat{\delta}(q_2,\varepsilon),0)$.

=ε-closure($\delta(\{q_0q_1q_2\},0)$.

=ε-closure($\delta(q_0,0)U\delta(q_1,0)U\delta(q_2,0)$).

=ε-closure($q_0Uq_2U\phi$).

=ε-closure(q_0)Uε-closure(q_2)Uε-closure(ϕ).

=$\{q_0q_1\}U\{q_0q_1q_2\}U(\phi)$

=$\{q_0q_1q_2\}$

$\hat{\delta}(q_2,1)$=ε-closure($\delta(\hat{\delta}(q_2,\varepsilon),1)$.

=ε-closure($\delta(\{q_0q_1q_2\},1)$.

=ε-closure($\delta(q_0,1)U\delta(q_1,1)U\delta(q_2,1)$).

=ε-closure($q_0U\phi U\phi$).

=ε-closure(q_0)Uε-closure(ϕ)Uε-closure(ϕ)

=(q_0q_1)U(ϕ)U(ϕ)

=$\{q_0q_1\}$

Table1.5:Equivalencetableofε-NFAandNFA.

Q \ Σ	0	1
→q0	{q0q1q2}	{q0q1 }
q1	{q0q1q2}	φ
*q2	{q0q1q2}	{q0q1}

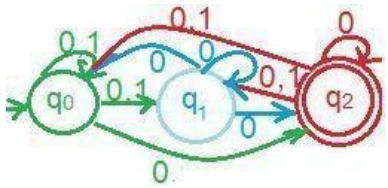


Figure1.58:NFAwithoutε-transition.

Conversion of ε-NFA to DFA:

ItissameasconversionofNFAtoDFA,beginswithinitialstateofε-NFA.

Example1.23: Convertthe ε-NFAshowninfigure1.59intoDFA.

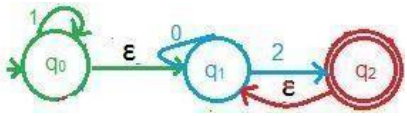


Figure1.59:ε-NFAofExample1.23.

Firstfindoutthe ε-closureofallthestates

$\hat{\delta}(q_0, \varepsilon)$ = ε-closure(q_0)= {q_0q_1}
$\hat{\delta}(q_1, \varepsilon)$ = ε-closure(q_1)= q_1
$\hat{\delta}(q_2, \varepsilon)$ = ε-closure(q_2) = {q_1q_2}

Table1.6:Equivalencetableofε-NFAandNFA.

Q \ Σ	0	1	2
→q0	q1	[q0q1]	[q1q2]
q1	q1	φ	[q1q2]
[q0 q1]	q1	[q0 q1]	[q1 q2]
*[q1q2]	q1	φ	[q1 q2]

$\hat{\delta}$ (q_0,0)=ε-closure(δ($\hat{\delta}$ (q_0,ε),0).

=ε-closure(δ({q_0q_1},0).

$=\varepsilon\text{-closure}(\delta(q_0,0)U\delta(q_1,0)).$

$=\varepsilon\text{-closure}(\phi Uq_1).$

$=\varepsilon\text{-closure}(\phi)U\varepsilon\text{-closure}(q_1).$

$=\phi U\ q_1$

$=q_1$

$\hat{\delta}(q_0,1)=\varepsilon\text{-closure}(\delta(\hat{\delta}(q_0,\varepsilon),1).$

$=\varepsilon\text{-closure}(\delta(\{q_0q_1\},1).$

$=\varepsilon\text{-closure}(\delta(q_0,1)U\delta(q_1,1)).$

$=\varepsilon\text{-closure}(q_0U\phi).$

$=\varepsilon\text{-closure}(q_0)U\varepsilon\text{-closure}(\phi).$

$=\{q_0q_1\}U\phi$

$=[q_0q_1]$

$\hat{\delta}(q_0,2)=\varepsilon\text{-closure}(\delta(\hat{\delta}(q_0,\varepsilon),2).$

$=\varepsilon\text{-closure}(\delta(\{q_0q_1\},2).$

$=\varepsilon\text{-closure}(\delta(q_0,2)U\delta(q_1,2)).$

$=\varepsilon\text{-closure}(\phi Uq_2).$

$=\varepsilon\text{-closure}(q_2).$

$=[q_1q_2]$

$\hat{\delta}(q_1,0)=\varepsilon\text{-closure}(\delta(\hat{\delta}(q_1,\varepsilon),0).$

$=\varepsilon\text{-closure}(\delta(q_1,0).$

$=\varepsilon\text{-closure}(q_1).$

$=q_1$

$\hat{\delta}(q_1,1)=\varepsilon\text{-closure}(\delta(\hat{\delta}(q_1,\varepsilon),1).$

$=\varepsilon\text{-closure}(\delta(q_1,1).$

$=\varepsilon\text{-closure}(\phi).$

$=\phi$

$\hat{\delta}(q_1,2)=\varepsilon\text{-closure}(\delta(\hat{\delta}(q_1,\varepsilon),2).$

$=\varepsilon\text{-closure}(\delta(q_1,2)$

$=\varepsilon\text{-closure}(q_2)$

$=[q_1q_2]$

$\hat{\delta}([q_0q_1],0)=\varepsilon\text{-closure}(\delta(\hat{\delta}([q_0q_1],\varepsilon),0)$

$=\varepsilon\text{-closure}(\delta(\hat{\delta}(q_0,\varepsilon)U\hat{\delta}(q_1,\varepsilon)),0).$

$=\varepsilon\text{-closure}(\delta((q_0,q_1)Uq_1),0)).$

$=\varepsilon\text{-closure}(\delta((q_0,q_1),0)).$

$=\varepsilon\text{-closure}(\delta(q_0,0)U\delta(q_1,0)).$

$=\varepsilon\text{-closure}(\phi Uq_1)$

$=\varepsilon\text{-closure}(q_1)$

$=q_1$

$\hat{\delta}([q_0q_1],1)=\varepsilon\text{-closure}(\delta(\hat{\delta}([q_0q_1],\varepsilon),1)$

$=\varepsilon\text{-closure}(\delta(\hat{\delta}(q_0,\varepsilon)U\hat{\delta}(q_1,\varepsilon)),1).$

$=\varepsilon\text{-closure}(\delta((q_0,q_1)Uq_1),1)).$

$=\varepsilon\text{-closure}(\delta((q_0,q_1),1)).$

$=\varepsilon\text{-closure}(\delta(q_0,1)U\delta(q_1,1)).$

$=\varepsilon\text{-closure}(q_0U\phi)$

$=\varepsilon\text{-closure}(q_0)$

$=[q_0,q_1]$

$\hat{\delta}([q_0q_1],2)=\varepsilon\text{-closure}(\delta(\hat{\delta}([q_0q_1],\varepsilon),2)$

$=\varepsilon\text{-closure}(\delta(\hat{\delta}(q_0,\varepsilon)U\hat{\delta}(q_1,\varepsilon)),2).$

$=\varepsilon\text{-closure}(\delta((q_0,q_1)Uq_1),2)).$

$=\varepsilon\text{-closure}(\delta((q_0,q_1),2)).$

$=\varepsilon\text{-closure}(\delta(q_0,2)U\delta(q_1,2)).$

=ε-closure(ϕUq_2)

=ε-closure(q_2)

=[q_1q_2]

$\hat{\delta}$([q_1q_2],0)=ε-closure(δ($\hat{\delta}$([q_1q_2],ε),0)

=ε-closure(δ($\hat{\delta}$(q_1,ε)U$\hat{\delta}$(q_2,ε),0)).

=[q_1q_2]

Table1.7:Equivalencetableofε-NFAandDFA.

Q / Σ	0	1	2
→q_0	q_1	[q_0q_1]	[q_1q_2]
q_1	q_1	φ	[q_1q_2]
[q_0 q_1]	q_1	[q_0 q_1]	[q_1 q_2]
[q_1q_2]	q_1	φ	[q_1 q_2]
φ	φ	φ	φ

$\hat{\delta}$(ϕ,(0,1,2))=ϕ

Markthefinalstates,any pairwhichhasafinalstateofε-NFAtheparticularpairisafinalstateinDFA.

Table 1.7: Equivalence table of ε-NFA and DFA .

Q / Σ	0	1	2
→q_0	q_1	[q_0q_1]	[q_1q_2]
q_1	q_1	φ	[q_1q_2]
[q_0 q_1]	q_1	[q_0 q_1]	[q_1 q_2]
*[q_1q_2]	q_1	φ	[q_1 q_2]
φ	φ	φ	φ

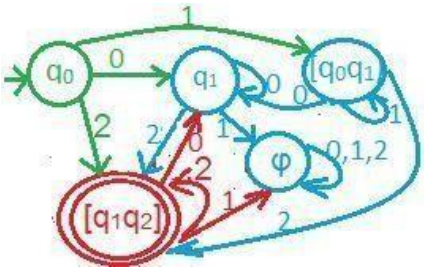


Figure1.60:DFAoffigure1.59.